

Huido Choi

M.S. Student · CLIM Lab (Global Climate Change, Innovative Monitoring & Modeling Lab)

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EDUCATION

M.S. in Climate & Environmental Science, Kookmin University Feb 2026 – Present

Dept. of Climate Technology Convergence, Graduate School · CLIM Lab (Global Climate Change, Innovative Monitoring & Modeling Lab) · Advisor: Prof. Chul-Hee Lim

B.S. in Forestry, Environment and Systems, Kookmin University Mar 2021 – Feb 2026

Minor: School of Software Convergence · Interdisciplinary Major: Climate Big Data

GPA 4.19 / 4.5 (Major 4.3 / 4.5) · 144 credits

Boseong High School, Seoul Mar 2018 – Feb 2021

RESEARCH INTERESTS

Climate change risk assessment (IPCC AR6 framework) · Explainable AI (XAI) and graph neural networks (GAT, GCN) · Climate justice and adaptation policy evaluation · Remote sensing-based forest and urban monitoring

PUBLICATIONS

Climate justice through explainable graph neural networks: A spatiotemporal attention-based urban heat risk assessment under IPCC AR6 framework 2026

Choi, H., Park, J. S., & Lim, C.-H. (2026) · Urban Climate, Article 102981 · First author

Impact Factor 6.9, CiteScore 12.2, top 6% in field · <https://doi.org/10.1016/j.uclim.2026.102981>

RESEARCH EXPERIENCE

Graduate Researcher (M.S. Student) · CLIM Lab, Kookmin University Feb 2026 – Present

Undergraduate Researcher · CLIM Lab, Kookmin University Feb 2025 – Feb 2026

Winter Research Intern · CLIM Lab, Kookmin University Dec 2024 – Jan 2025

RESEARCH PROJECTS

Funded R&D Projects (Research Assistant / Participating Researcher)

1. Inequality and the Value of Adaptation in the Era of Climate Crisis: A Spatial Approach for Climate Justice

Ministry of Education · National Research Foundation of Korea | Research Assistant | Mar 2025 – Apr 2025

2. Development of a Digital Innovation-Based Policy Platform for Mental Health Resilience in the Era of Climate Crisis

National Research Foundation of Korea | Research Assistant | Jun 2025 – Mar 2026

3. Development of an Integrated Forest Assessment Model Based on AI and Climate Technology-Policy Convergence

Ministry of Science and ICT (Outstanding Young Researcher Program) | Research Assistant | May 2025 – Feb 2026

4. Optimal Access/Evacuation Route Analysis for Wildfire Suppression Resources and Stakeholder-Tailored Information Sharing Technology

Korea Forest Service · Korea Forestry Promotion Institute | Research Assistant | Jul 2025 – Feb 2026

5. Field-Tailored Workforce Training for the Forest Industry (R&D): Integrated Landslide Management Specialists

Korea Forest Service | Research Assistant | Jan 2026 – Dec 2026

6. Integrated Landslide Management Specialist Training Center

Korea Forest Service | Participating Researcher | Jan 2026 – Present

7. Development of AI-Based Forest Road Route Selection and Design Technology

Korea Forest Service · Korea Forestry Promotion Institute | Research Assistant | Apr 2026 – Present

8. Development of Ecosystem Restoration Technologies for Enhancing Ecosystem Value

Ministry of Climate, Energy and Environment · Korea Environmental Industry & Technology Institute | Research Assistant | Apr 2026 – Present

9. Technology for Analyzing Corporate Financial Impacts of Physical and Transition Climate Risks

Ministry of Climate, Energy and Environment · Korea Environmental Industry & Technology Institute | Research Assistant | Apr 2026 – Present

10. Kookmin University Carbon Sink Specialized Graduate School Program

Korea Forestry Promotion Institute | Participating Researcher | Jun 2026 – Present

Commissioned Research Projects

1. Translation of UNFCCC Expert Training Courses

National Institute of Forest Science | Research Assistant | Oct 2024 – Nov 2024

2. Analysis of Support Systems for Overseas Forest Resource Development

Korea Forestry Promotion Institute | Research Assistant | Oct 2025 – Dec 2025

3. Training of Bioclimatic Research Specialists (2026)

National Institute of Biological Resources | Principal Investigator | Apr 2026 – Dec 2026

4. Climate Risk Assessment under Compound Disasters (2026)

National Institute of Environmental Research | Research Assistant | Jun 2026 – Dec 2026

CONFERENCE PRESENTATIONS

AI-Driven Climate-Resilient Adaptation Model of Korean Forests (CRAFT): Grid-scale Integration of Growth, Carbon, Habitat, and Disaster

Aug 2026

Joint International Symposium on Sustainable Forest Ecosystem Management 2026 (SFEM 2026), Daegu, Republic of Korea | Oral, first author

A Multi-Objective Optimization Framework for Forest Road Network Installation Prioritization: A Pareto Frontier Approach

Aug 2026

2026 Summer Conference of the Korean Society of Forest Science, Yeosu (in Korean) | Poster, second author

Impact of Climate Change on Burned Area Trends: Focusing on the Indochinese Peninsula

2026

KSSCCR 2026 Spring Conference (in Korean) | Poster, second author

Mortality Reduction Effects and Optimal Allocation of Heatwave Response Budgets in Seoul

2026

KSSCCR 2026 Spring Conference (in Korean) | Oral, first author · Focusing on the 25 districts of Seoul

Resilience Analysis of Forest Ecosystems in Gangwon Province: EWS-Based Time-Series Analysis with AR(1) Indicators

Feb 2026

2026 Joint Conference of Korean Forest Science Societies (in Korean) | Poster, second author

Explainable AI (XAI) for Climate Justice: GAT- and AR6 Framework-Based Spatiotemporal Heat Risk Assessment of Seoul	Dec 2025
KSCCR 2025 Fall Conference, Outstanding Graduate Student Program (in Korean) Oral · Prize Winner	
Eco-Community Center Development Plan Utilizing Vacant Houses in Jeongneung 3-dong	Nov 2025
2025 HUSS Environmental Consortium Conference "Climate Future" (in Korean) Oral, first author	
Cost and Effectiveness Evaluation of Adaptation for Mitigating Climate Inequality and Flood Risk: Focusing on Districts of Seoul	Jun 2025
KSCCR 2025 Spring Conference (in Korean) Oral, second author · Best Presentation Paper Award	
Urban Heat Island Risk Assessment Using Spatiotemporal GCN	Jun 2025
Korean Society of Climate Change Research (KSCCR) 2025 Spring Conference (in Korean) Poster, first author · Best Poster Paper Award	

HONORS & AWARDS

Grand Prize (AI Service Creation Award) · 1st Forest Science AI Competition, National Institute of Forest Science	Jul 2026
Multi-objective forest management decision-support AI (MOFOM) · Team lead · ₩2,000,000	
Excellence Award · 2026 MCEE AX Idea Competition (KEPCO KDN · Korea Southeast Power)	Jul 2026
Quadruple verification of greenhouse-gas disclosure using satellite imagery · ₩3,000,000	
Excellence Award (KHTC Chairman's Award) · 2026 Forest Public Data & AI Startup Competition, Korea Forest Service	Jul 2026
Soop Starter — settlement decision support for young foresters · Team lead	
Excellence Award (Korea National Park Service Chairman's Award) · 2025 Environmental Data Utilization & Analysis Contest	2025
Climate vulnerability prediction for Seoul using GAT-LSTM with climate scenarios · ₩2,500,000	
Prize, Outstanding Graduate Student Support Program · KSCCR 2025 Fall Conference	Dec 2025
Explainable AI for climate justice: GAT- and AR6-based spatiotemporal heat risk assessment of Seoul	
Encouragement Award · 2025 Kookmin University Climate Change Response Business Idea Contest	Dec 2025
Climate adaptation platform "PCAR" financial product	
Excellence Award · 2025 Kookmin University Interdisciplinary Major Essay Contest	Dec 2025
"Learning Climate Big Data"	
Best Poster Paper Award · KSCCR 2025 Spring Conference	Jun 2025
Urban heat island risk assessment for Seoul using a spatiotemporal GCN · First author	
Best Presentation Paper Award (second author) · KSCCR 2025 Spring Conference	Jun 2025
Cost and effectiveness of adaptation for mitigating climate inequality and flood risk in Seoul	
Outstanding University Climate Activist · Climate Change Center U-SAVERS	Nov 2024

SCHOLARSHIPS

Forest Pioneer Research Fellow, 5th cohort · Chung In-wook Academic Scholarship Foundation (Sampyo Group)	2026 – 2028
Competitive research fellowship for graduate researchers developing forest and forestry technology applicable directly in the field · five researchers selected nationwide · KRW 1,000,000 per month for up to 24 months (up to KRW 24,000,000) · reviewed by the Korean Society of Forest Science	
· Seonggok Scholarship · Kookmin University Graduate School	2025 – 2027
· Top Student Scholarship (70% of tuition) · Kookmin University	2024 – 2025

CERTIFICATIONS

· Preparing a Greenhouse Gas Inventory under the Enhanced Transparency Framework FAO eLearning Academy	Apr 2026
· Engineer Greenhouse Gas Management · Human Resources Development Service of Korea	Dec 2025
· Engineer Big Data Analysis · Korea Data Agency (K-DATA)	Dec 2025
· Industrial Engineer Forest · Human Resources Development Service of Korea	Sep 2025
· ADsP (Advanced Data Analytics Semi-Professional) · Korea Data Agency (K-DATA)	Mar 2025
· Computer Specialist in Spreadsheet & Database, Level 1 · KCCI	Sep 2024
· Korean History Proficiency Test, Level 1 · National Institute of Korean History	Dec 2023
· Ultra-light Vehicle (Drone) Pilot, Class 4 · Korea Transportation Safety Authority	Dec 2022

TECHNICAL SKILLS

Programming & Analysis Python (PyTorch), R

AI / ML GNN (GAT, GCN), LSTM, XAI (SHAP, GNNExplainer, Integrated Gradients)

Geospatial & Remote Sensing ArcGIS Pro, Google Earth Engine, satellite image analysis

Web & Visualization HTML/CSS/JavaScript, React, Leaflet.js, D3.js, Chart.js

Frameworks IPCC AR6 risk assessment framework, SSP scenarios

SELECTED PROJECTS

AI services and analytical systems

- Multi-objective Forest Management Decision-Support AI (MOFOM) Grand Prize, Forest Science AI Competition
<https://mofom-ai.vercel.app>
<https://github.com/jwn6174-crypto/forest-ai-agent>
- Soop Starter — settlement decision support for young foresters Excellence Award, KFS Startup Competition
<https://soop-starter.streamlit.app>
<https://github.com/zxsa0716/soop-starter>
- Quadruple verification of GHG disclosure via satellite imagery Excellence Award, AX Idea Competition
https://github.com/zxsa0716/AX_Contest
- Seoul climate vulnerability prediction (GAT-LSTM) Excellence Award, Environmental Data Contest
<https://www.youtube.com/watch?v=jxCs3xLc0wY>
- CINA — Climate Issue-Network Analysis platform
<https://zxsa0716.github.io/cina/web/index.html>
<https://github.com/zxsa0716/cina>
- Seoul commercial district risk analysis platform
<https://seoul-commercial-district-risk.netlify.app>

Web services and data visualisation

- Personal research portfolio website
<https://heedo-choi.vercel.app>
- Seoul climate change vulnerability web map
https://zxsa0716.github.io/vulnerability_seoul/seoul_vulnerability_dashboard.html
- Seoul youth single-person household safety-net map
<https://seoul-youth-platform.netlify.app>
- Website for Dept. of Climate Technology Convergence, Kookmin University
<https://cms.kookmin.ac.kr/climtech/index.do>
- Kookmin University's first 2024 Sustainability Report (Climate Crisis & ESG Management)

<https://cms.kookmin.ac.kr/sustainability>

- Community web game for reducing illegal dumping in Seongbuk-gu
<https://eco-quest-app.netlify.app>
- Technical blog on spatial analysis and climate data
<https://zxsa716.tistory.com>

EXTRACURRICULAR ACTIVITIES

President, 'Greenery' Academic Society, Kookmin University	Jan 2025 – Dec 2025
Dept. of Forestry, Environment and Systems · Seongbuk-gu Community Building Project (selected & executed) · 2025 Seoul Youth Volunteer Planning Corps (2nd cohort) · Community lecture, Seongbuk Village Community Program	
U-SAVERS University Climate Activist, Climate Change Center	Mar 2024 – Nov 2024
Climate Change Young Leadership Academy (11th) · Plastic School Attack program leader · Outstanding Activist Award	
'Green Narae' Youth Environmental Education Club, Mullae Youth Center	Feb 2024 – Dec 2024
Developed and delivered environmental education for youth with developmental disabilities · Main MC, 2024 Seoul Youth Environmental Festival	
Sparta Coding Club 'Letanz' Supporters	Jan 2022 – Feb 2022

TRAINING

· Forest Big Data Analysis & Utilization Training (Korea Forestry Promotion Institute, 15 hrs)	Feb 2024
· Climate Change Young Leadership Academy, 11th (Climate Change Center, 32 hrs)	Mar – Nov 2024
· Python Practical Training (24 hrs)	Jul – Dec 2022
· Sparta Coding Club Web Development (40 hrs)	Jan – Feb 2022

VOLUNTEER EXPERIENCE

Organisation	Activity	Hours
Mullae Youth Center	Ran the youth environmental-education club "Greennarae" (Feb-Dec 2024)	78
Gayang 7 Community Welfare Center	Community welfare service	64
Climate Change Center	U-SAVERS university climate activist, 4th cohort (Mar-Nov 2024)	23
Seoul Volunteer Center	2025 Seoul Youth Planning Volunteer Corps, 2nd cohort (May-Sep 2025)	13
Hangang Project HQ, Banpo Center	Visitor guidance, Hangang Park	2
Total	5 organisations	180

MILITARY SERVICE

Sergeant, CBRN Battalion, 75th Division, Republic of Korea Army	Jul 2022 – Jan 2024
Honorably discharged after full service	